DSCI 552 Homework 4

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2. Time Series Classification Part 2: Binary and Multiclass Classification

I reused the HW3 feature-extraction code and checked its values against my HW3 GitHub repository. Part 1 answers are not resubmitted. All 88 original recordings are read so segmentation uses the raw time series, not an already summarized feature table.

The fixed split gives 69 training and 19 test recordings. Datasets 1-2 are test data for bending1/bending2; datasets 1-3 are test data for the other activities. Thus test count = 2 + 2 + 5(3) = 19 and training count = (7 - 2) + (6 - 2) + 5(15 - 3) = 69.

activity	train	test
bending1	5	2
bending2	4	2
cycling	12	3
lying	12	3
sitting	12	3
standing	12	3
walking	12	3

bending2/dataset4.csv uses whitespace, and cycling/dataset9.csv and dataset14.csv each have one extra trailing comma. No observed values are discarded. sitting/dataset8.csv has 479 rows because timestamp 13,500 ms is absent in the original ZIP; it is not imputed. The other 87 recordings have 480 rows.

Each recording is divided into l consecutive, nonoverlapping parts using np.array_split, retaining all observations. There are 6l subseries and 42l features (seven statistics per subseries), with 88 recordings for every l. Standard deviation uses ddof=1 and quartiles use linear interpolation. Feature names identify original channel and segment, e.g. mean_s1_p2.

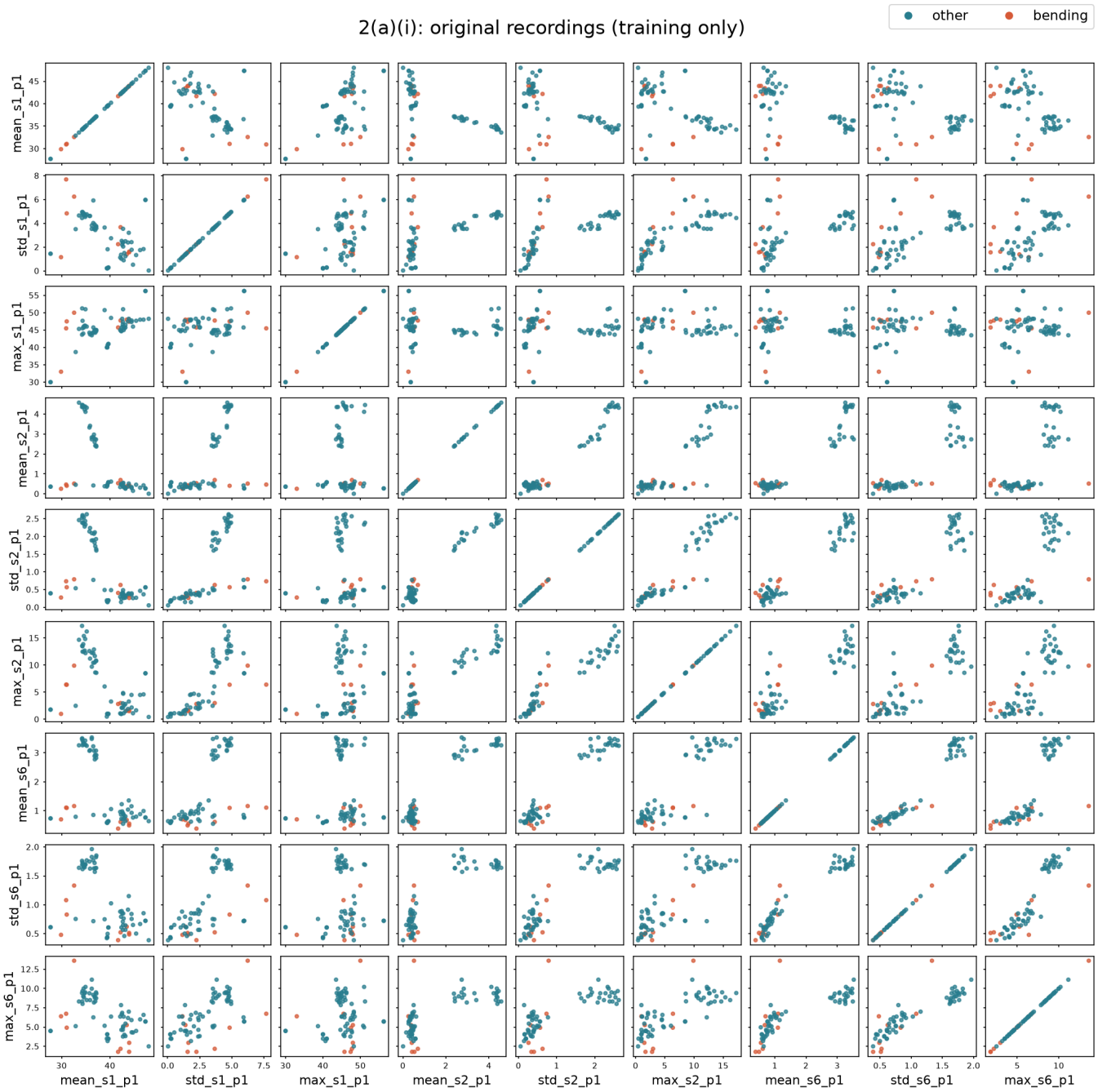
2(a)(i) Original-series scatter plots

Bending1 and bending2 are combined as bending (positive class). I plotted the HW3-selected mean, standard deviation, and maximum from original channels 1, 2, and 6, using only the 69 training recordings. There are nine features and nine bending versus 60 other training cases. The next page shows all pairwise scatter plots.

feature	class	min	max	mean
mean_s1_p1	other	27.7164	48.0042	39.1945
mean_s1_p1	bending	29.8819	43.9691	37.7446
std_s2_p1	other	0.0551	2.6255	1.0848
std_s2_p1	bending	0.2631	0.7900	0.4801
max_s6_p1	other	2.5000	11.1500	6.7038
max_s6_p1	bending	1.7900	13.6100	5.0711

Mean channel 1 overlaps substantially: the bending range lies within the other range. Bending has lower average std channel 2, but the ranges overlap. Some pairs put bending in a low-variation region; the plots do not give a clean one-feature cutoff.

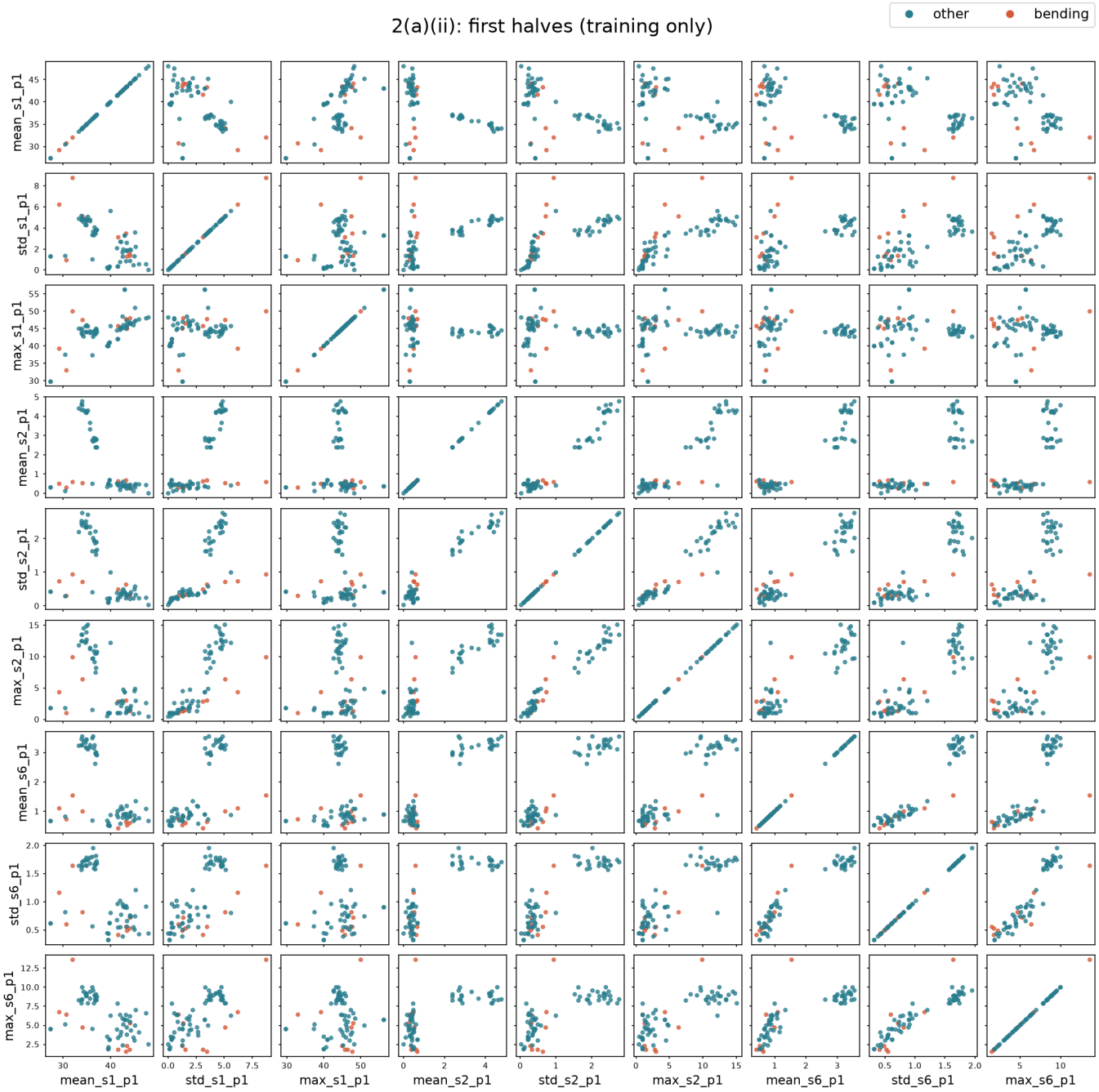
2(a)(i) Scatter matrix: original recordings



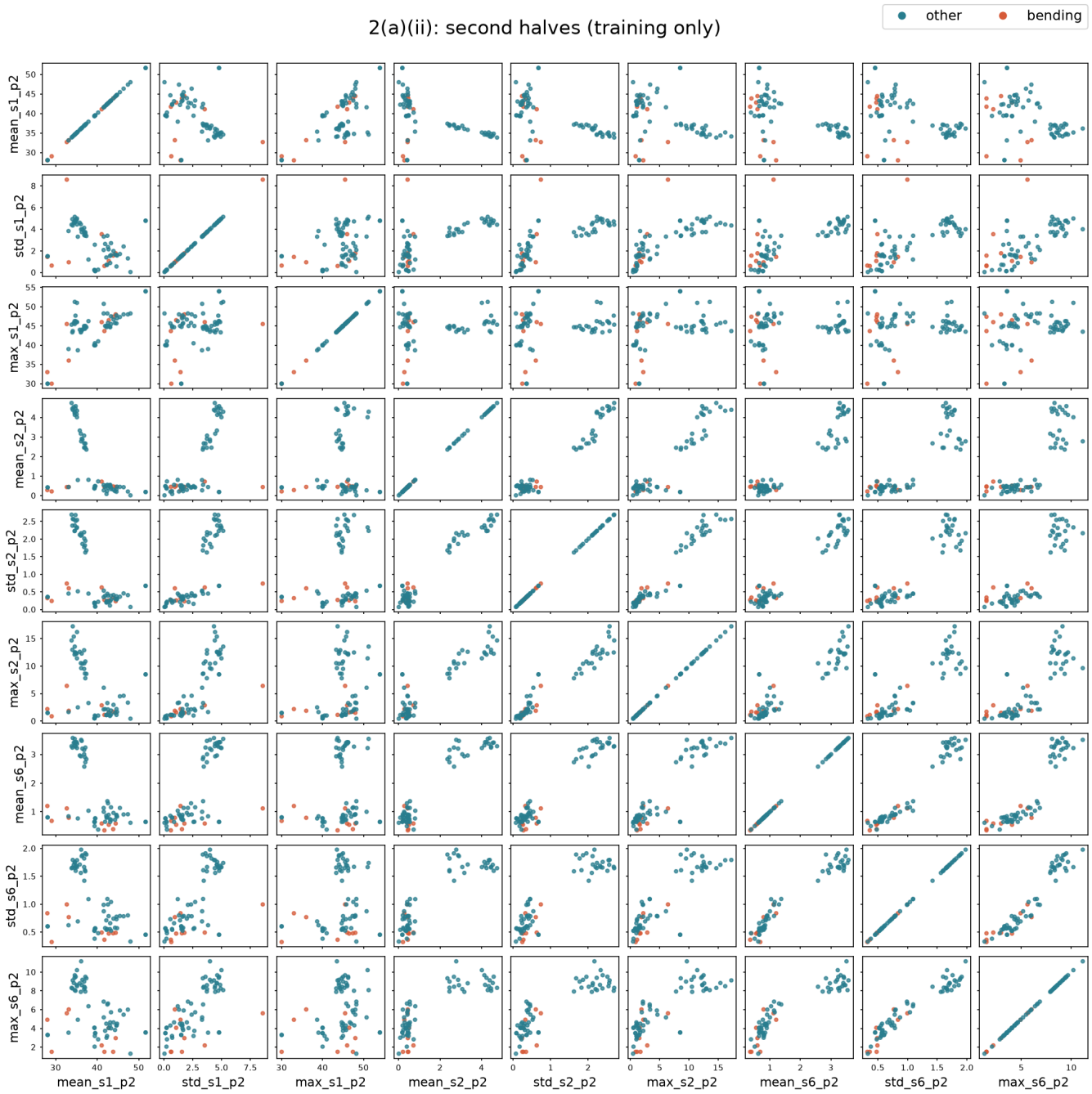
Orange = bending; blue = other. Axes use the feature names described on the previous page. Diagonal panels repeat a feature against itself.

2(a)(ii) Split-series scatter plots: first half

Splitting into two parts creates 12 subseries and 84 total features per recording; the training sample remains 69. The requested channels give 18 plotted features. The following three blocks show within-first-half, within-second-half, and all cross-half pairs. A transposed cross block would repeat the same pairs.

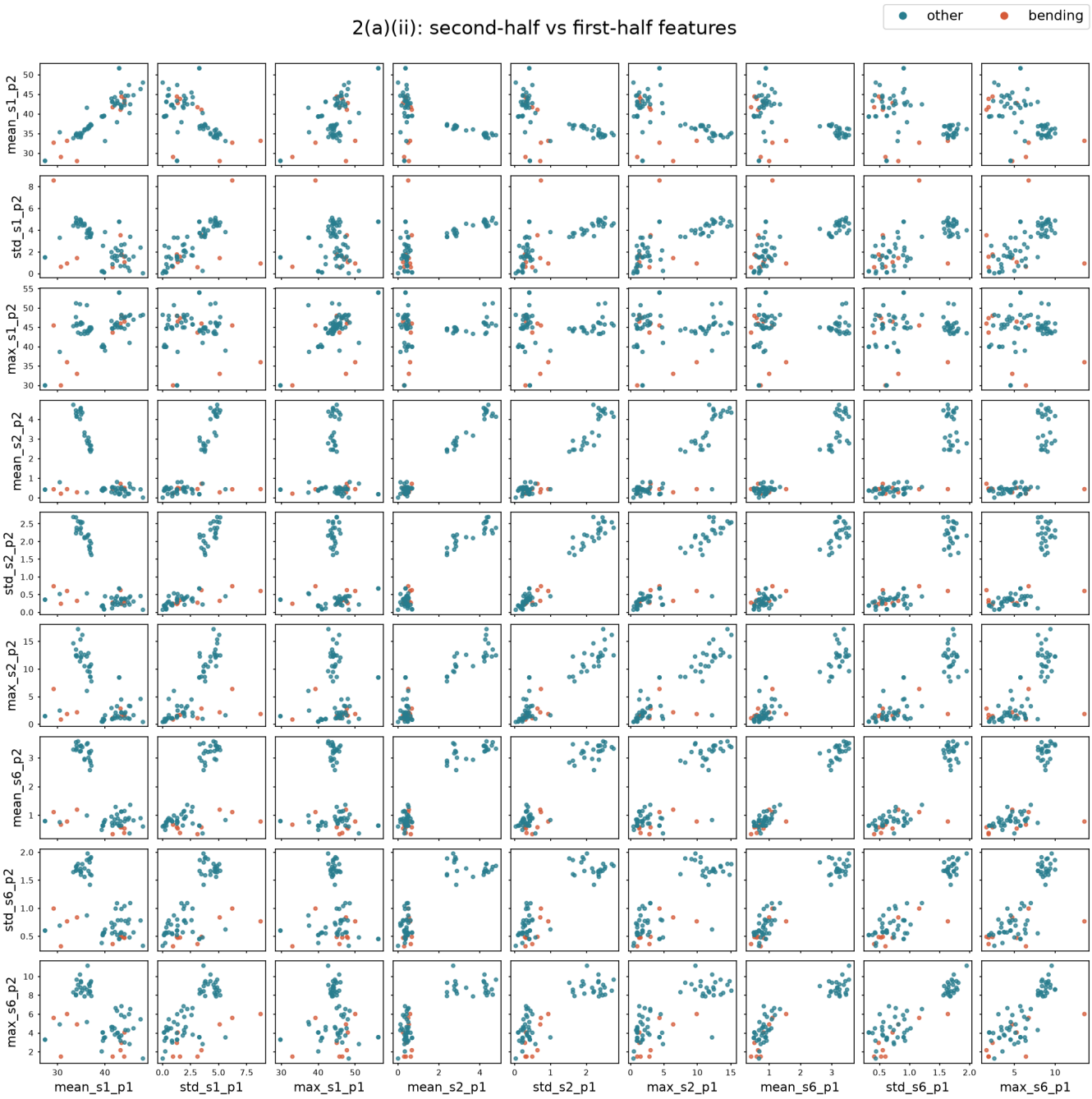


2(a)(ii) Split-series scatter plots: second half



These are features from the second half of the same recordings, not additional independent activity instances.

2(a)(ii) Scatter matrix: cross-half comparisons



The broad pattern is similar before and after splitting. Some points move or spread out, but there is no new clean separation across all panels. Shorter windows can reveal changes during an activity, while extrema and standard deviations become more sensitive to the window. Both halves still belong to the same recording.

2(a)(iii) Ordinary logistic regression, RFE, and CV

I used RFE, the assignment's alternative to p-value pruning. After standardization, it removes features with the smallest squared logistic coefficients. RFE ranks are not statistical p-values. All seven statistics from all six channels and all l parts are eligible. C = infinity removes the penalty from the lbfgs logistic model; tolerance is 1e-6 and max_iter is 2,000.

For every l = 1,...,20, the searched p values are 1, 2, 3, 4, 5, 6, 8, 10, 12, 18, 24, 36, 54, 84, omitting p greater than 42l. RFE removes 25% of the original feature count per step until it reaches p. The result is best on this stated grid, not an exhaustive search over all possible subsets.

Five-fold StratifiedKFold uses shuffling and seed 552. With only nine bending cases, stratification keeps both classes in each fitting and validation part. Scaling, RFE, and the classifier are all fitted inside each fold through a pipeline. Segmentation has no fitted parameters, so it is safe to compute it beforehand.

Correct CV: fit every learned step on the fitting fold, then score untouched validation observations. **Wrong CV:** scale or select features using all training observations before splitting into folds. Then validation observations affect the selected representation and the score is optimistic. Test observations never choose l, p, C, or thresholds.

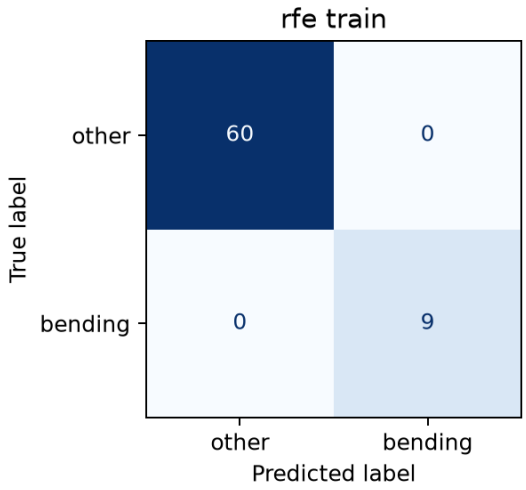
The objective is mean fold accuracy. Ties prefer smaller l and then smaller p (or smaller C for regularized models). The CV maximum is a tuning score and can be optimistic after many comparisons. Balanced accuracy is also reported at evaluation.

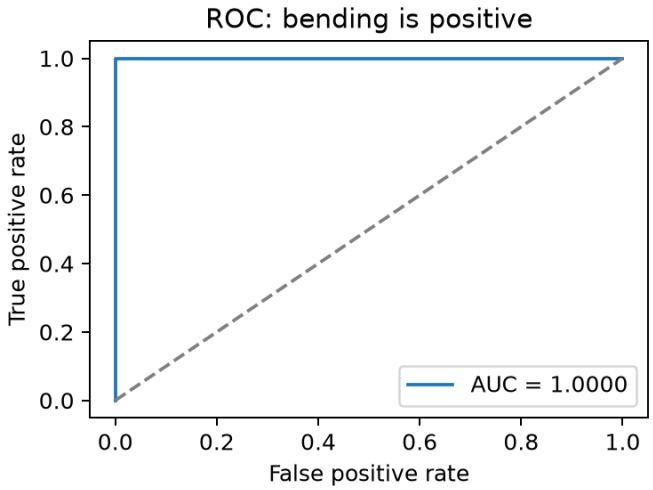
l	p	CV accuracy	CV SD
1	4	0.9857	0.0286
2	2	0.9571	0.0350
3	3	0.9714	0.0350
4	24	0.9429	0.0700
5	5	0.9418	0.0835
6	3	0.9429	0.0535
7	3	0.9286	0.0452
8	10	0.9571	0.0571
9	4	0.9714	0.0350
10	18	0.9429	0.0700
11	18	0.9571	0.0571
12	8	0.9571	0.0571
13	36	0.9143	0.1143
14	12	0.9286	0.0639
15	10	0.9132	0.0532
16	4	0.9286	0.0639
17	10	0.8989	0.0724
18	5	0.9714	0.0350
19	5	0.9714	0.0350
20	10	0.9714	0.0571

Selected l = 1, p = 4; mean CV accuracy = 0.9857. Full search scores are in results/rfe_cv.csv.

2(a)(iv) Training evaluation, coefficients, and p-values

The selected pipeline is refitted using all 69 training recordings. Bending is positive, the decision threshold is 0.5, and confusion-matrix rows are true labels, columns predicted labels. ROC uses the predicted bending probability.





term	beta (scaled)	beta (raw)	nominal Wald p	valid p?
intercept	-34.4095	-11.6817	0.9379	False
q3_s1_p1	-13.4217	-2.8263	0.9870	False
max_s2_p1	-8.1463	-1.5941	0.9919	False
min_s5_p1	17.1474	3.2173	0.9580	False
max_s5_p1	20.4062	3.7719	0.9547	False

For a finite full-rank ordinary MLE, the Wald covariance is $(A^TWA)^{-1}$, W has diagonal p(1 - p), and the two-sided p-value is 2Φ(-|beta/SE|). The table shows the numerical calculation, not RFE ranks. The LP separation check below proves there is no finite ordinary MLE here: **these nominal Wald p-values are invalid**. Finite optimizer output is useful for prediction but does not establish finite likelihood estimates. A pseudoinverse is only a numerical diagnostic.

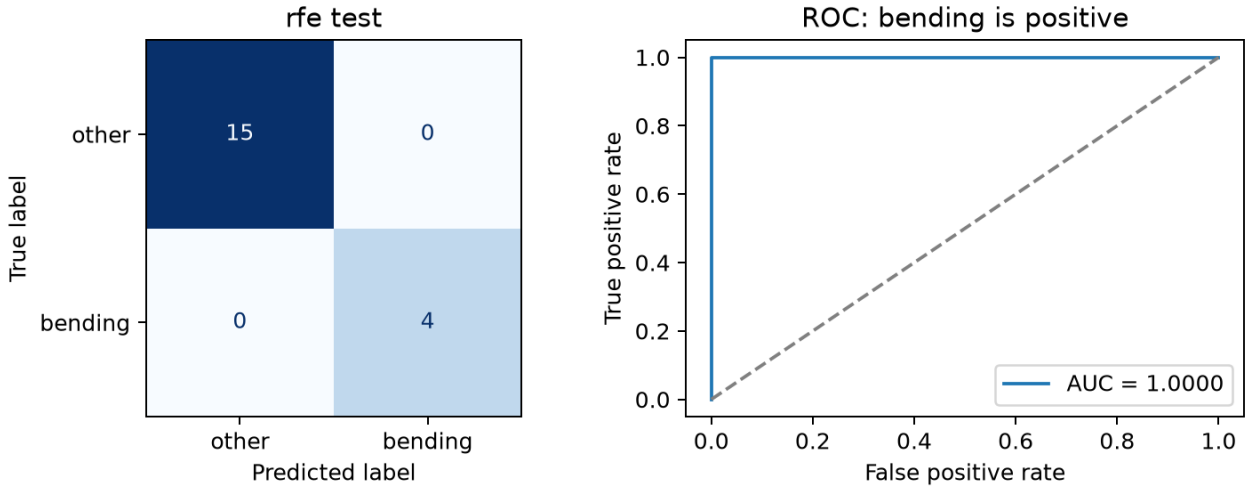
Supplementary finite-coefficient inference

term	Firth beta	penalized LR	approx. p
intercept	-3.9337	45.4567	1.560e-11
q3_s1_p1	-1.2470	2.9263	0.0871
max_s2_p1	-1.1010	1.7479	0.1861
min_s5_p1	1.4934	2.2403	0.1345
max_s5_p1	2.3686	17.8487	2.392e-05

Firth fitting maximizes log likelihood + 0.5 log determinant of the information matrix. For each test, fix one coefficient to zero, refit the others, and compare twice the penalized log-likelihood difference to chi-square(1). These approximate p-values refer to this separate bias-reduced fit, not the ordinary classifier, and are not adjusted for feature selection. Optimization gradients are checked in the notebook. [4]

2(a)(v) Test evaluation

I applied the selected l = 1, scaler, selected columns, and fitted coefficients to the 19 test recordings. No preprocessing or model parameters were refitted on test data. Test accuracy = 1.0000, compared with CV accuracy = 0.9857.



The test set is small: one prediction changes accuracy by 1/19 = 0.0526. Its activity composition also differs slightly from the training folds. The selected CV maximum has model-selection optimism. Therefore a difference between CV and test scores is expected and does not justify retuning on test labels.

The supplied data contain two exact train/test duplicates: cycling/dataset1.csv (test) equals cycling/dataset15.csv (train), and lying/dataset1.csv (test) equals lying/dataset15.csv (train). I retain the required split and all observations; therefore, 100% test accuracy on this split is not a guarantee of performance on fully independent new recordings.

2(a)(vi) Separation and coefficient instability

The complete-separation LP gives a strictly positive margin of 0.322438. Thus some coefficient vector gives the correct sign for every training observation. The information matrix has rank 5 of 5 and condition number 371.2253; full numerical rank does not eliminate separation.

tolerance	iterations	coefficient_norm	max_abs_coefficient	train_accuracy
0.0010	9	8.0230	5.5887	1.0000
1.000e-06	21	30.9346	20.4062	1.0000
1.000e-09	31	49.7919	32.4905	1.0000

Coefficient size increases as tolerance is tightened while training accuracy stays perfect. This is the behavior of separated logistic likelihood: coefficients can grow without a finite maximum. Solver stopping or early stopping, as suggested by the assignment, does not make ordinary Wald inference valid. The supplementary Firth analysis addresses finite-coefficient inference; the L1 model below addresses stability of prediction.

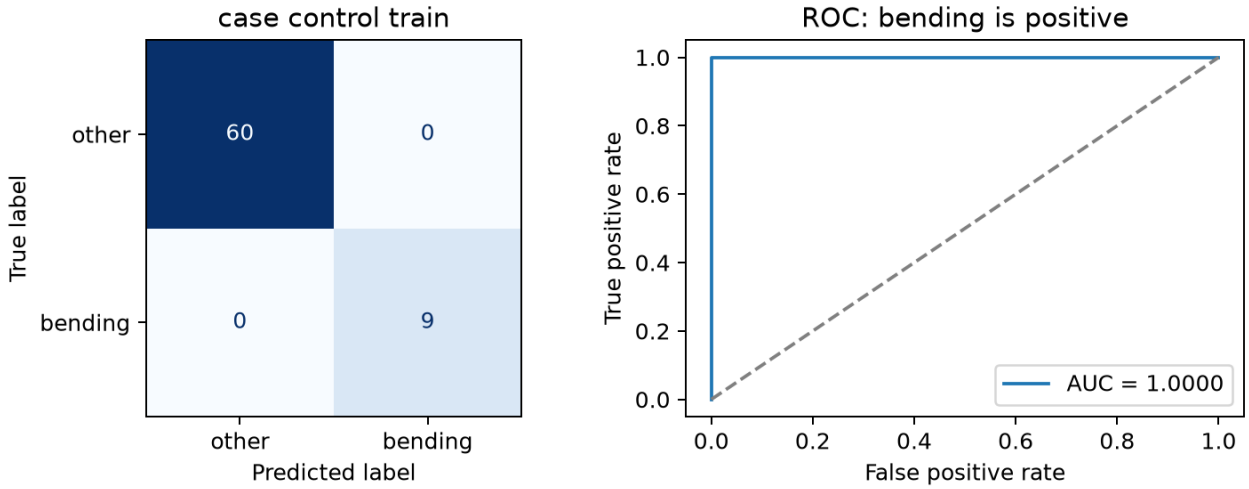
2(a)(vii) Case-control sampling and correction

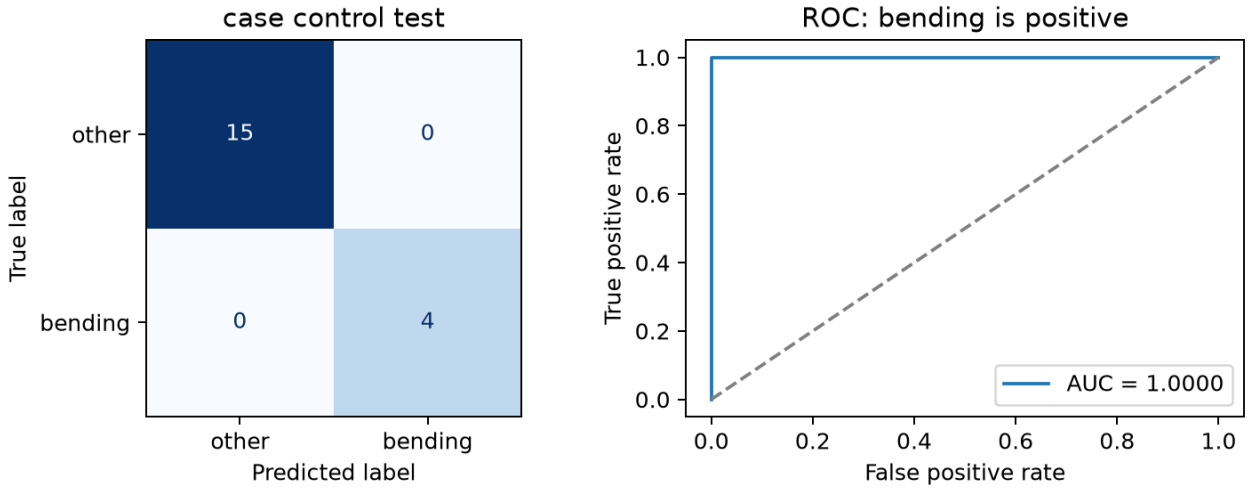
The counts show class imbalance even though the original confusion matrix is perfect: 60 other versus 9 bending training cases (and 15 versus 4 in test). I retained the 60 other cases and sampled 60 bending cases with replacement, using seed 552. The selected RFE features stay fixed.

$$\beta_{0,\,target} = \beta_{0,\,sample} + \log\frac{\pi}{1-\pi} - \log\frac{\pi_s}{1-\pi_s}$$

Target prevalence is the original training proportion 0.130435, while sampled prevalence is 0.500000. The intercept shift is -1.897120; the intercept changes from -33.319680 to -35.216800 in standardized coordinates. Slopes are unchanged by this correction. Test prevalence is not used.

Under outcome-only sampling, the correction restores the target prior odds; oversampling alone does not do that. Duplicate recordings add no independent observations. Separation still affects the fit, so the correction cannot guarantee probability calibration. Changing only the intercept cannot change score ordering or AUC.





Corrected training/test accuracies are 1.0000/1.0000; AUCs are 1.0000/1.0000. Here sampling and correction give no accuracy improvement over the original model, which already classified both sets perfectly.

2(b)(i) L1-penalized binary logistic regression

I searched every l = 1,...,20 and C = 0.01, 0.1, 1, 10, 100, 1000 using the same five binary folds. Standardization and L1 fitting are inside the pipeline. Liblinear fits the binary model; smaller C gives stronger shrinkage. Ties prefer smaller l, then smaller C.

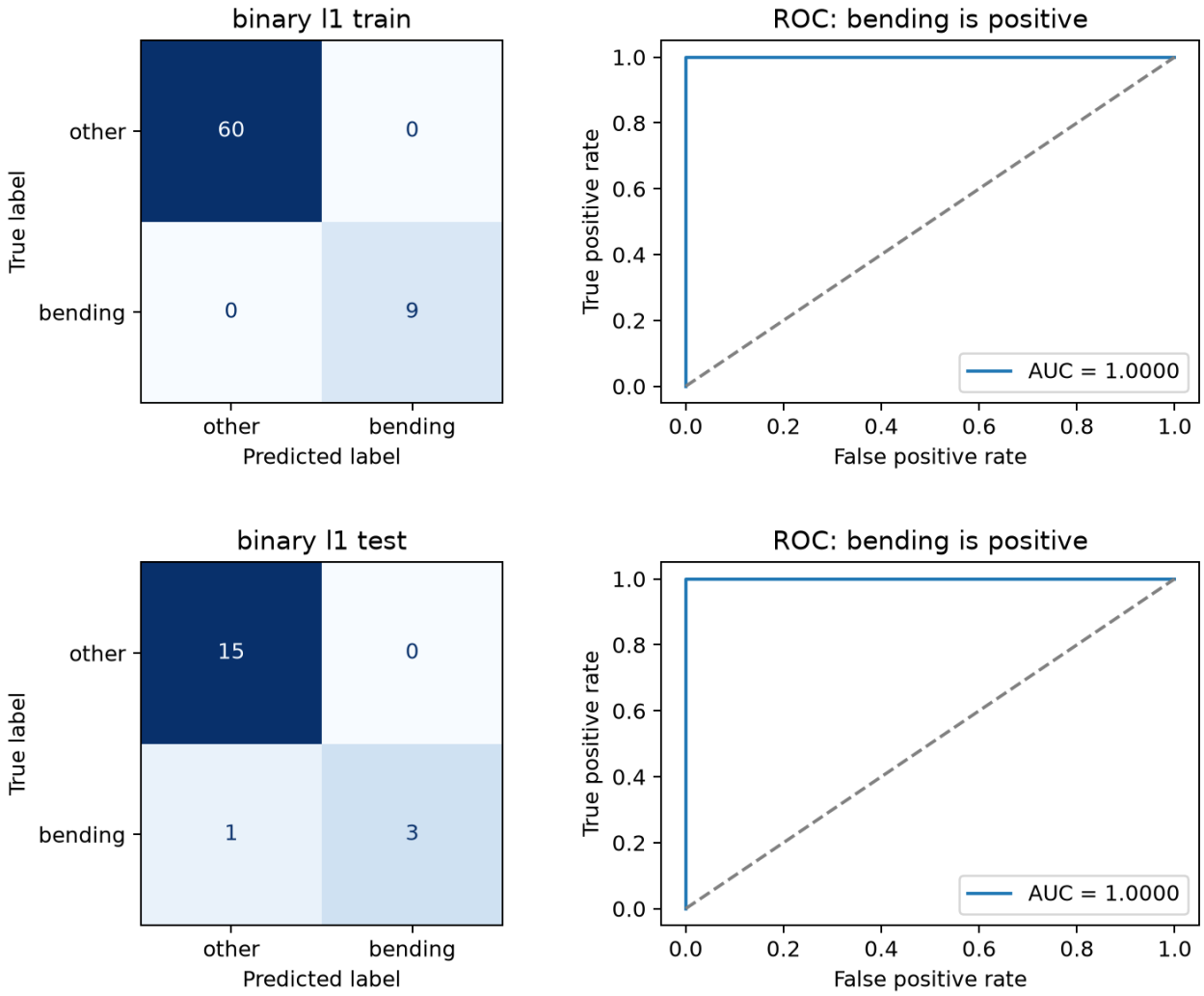
Selected l = 11, C = 1, with mean CV accuracy 0.9857 (SD 0.0286). There are 8 nonzero features at threshold 1e-8. No selected model or CV candidate emitted a convergence warning.

l	C	CV accuracy	CV SD
1	1.0000	0.9714	0.0350
2	1.0000	0.9429	0.0535
3	1.0000	0.9571	0.0350
4	1.0000	0.9571	0.0350
5	10.0000	0.9275	0.0452
6	1000.0000	0.9418	0.0538
7	1.0000	0.9286	0.0452
8	100.0000	0.9429	0.0535
9	1.0000	0.9286	0.0452
10	1000.0000	0.9286	0.0904
11	1.0000	0.9857	0.0286
12	1.0000	0.9857	0.0286
13	1.0000	0.9714	0.0350
14	1000.0000	0.9418	0.0538
15	1.0000	0.9429	0.0535
16	10.0000	0.9429	0.0535
17	1.0000	0.9429	0.0535
18	10.0000	0.9857	0.0286
19	100.0000	0.9857	0.0286
20	100.0000	0.9714	0.0571

Selected L1 features

feature	beta_standardized
max_s5_p1	0.2544
median_s5_p2	0.5809
q1_s5_p2	1.3110
max_s1_p6	-0.7862
median_s4_p8	-0.3091
std_s6_p9	-0.1466
min_s5_p10	0.7211
max_s6_p10	-0.0343

2(b)(i) Final binary L1 evaluation



2(b)(ii) L1 and the permitted RFE alternative

method	l	features	CV acc.	test acc.	test AUC
Unpenalized + RFE	1	4	0.9857	1.0000	1.0000
L1 logistic	11	8	0.9857	0.9474	1.0000

Question 2(a)(iii) permits RFE, which I chose because complete separation invalidates ordinary Wald p-values. Question 2(b)(ii) names p-value-based selection; **the measured comparison here is the permitted RFE alternative versus L1**, not a separate p-value-selected baseline.

P-value selection tests coefficients against a significance cutoff. RFE instead ranks coefficient importance and chooses the feature count by CV; its ranks are not p-values. L1 jointly shrinks and selects coefficients. Invalid Wald p-values cannot justify screening. The supplementary Firth tests describe the selected model; they were not used to select it.

Both models have CV accuracy 0.9857. RFE has test accuracy 1.0000 versus 0.9474 for L1 (one bending false negative); both test AUCs are 1.0000. RFE wins on this split; L1 is easier to implement and gives finite, sparse coefficients. These results cannot establish whether p-value-based selection would perform better.

2(c)(i) L1 multinomial logistic regression

The label has seven activities. SAGA fits a joint multinomial loss with L1 regularization, not seven independently fitted binary classifiers. I searched l = 1,...,20 and C = 0.01, 0.1, 1, 10, 100, 1000. StandardScaler is fitted inside each CV fitting fold.

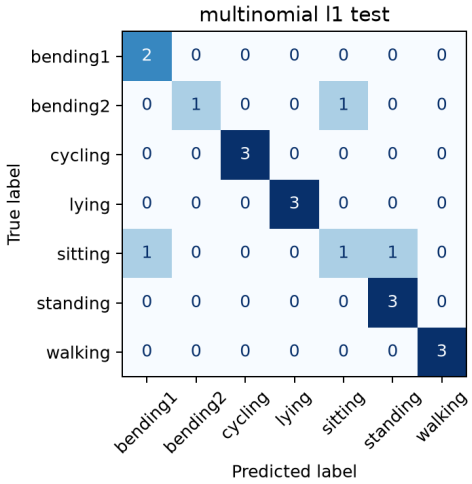
I use five stratified folds for all multiclass models, matching the assigned CV procedure. With only four bending2 training cases, one validation fold lacks bending2; every fitting fold still contains all seven activities. Accuracy remains defined in every fold. All models share these folds, seed 552, and the held-out test set. The saved fold-count table documents the missing validation class; fold-level multiclass ROC is not used.

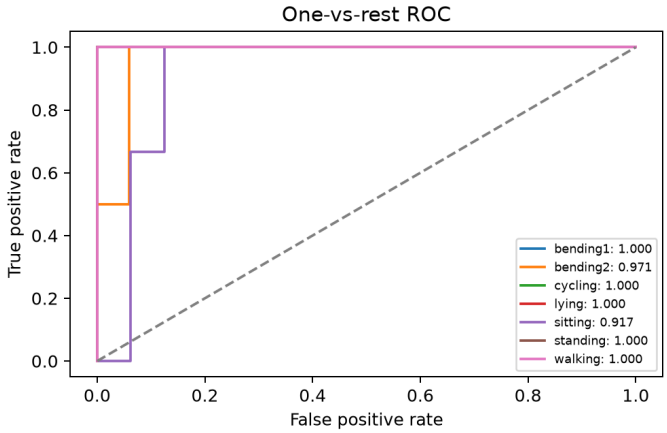
Selected l = 2, C = 1000. Mean CV accuracy = 0.9429 (SD 0.0286); 83 features have at least one nonzero class coefficient. Test accuracy = 0.8421; test error = 0.1579.

l	C	CV accuracy	CV SD
1	10.0000	0.9132	0.0281
2	1000.0000	0.9429	0.0286
3	100.0000	0.8835	0.0370
4	100.0000	0.8989	0.0342
5	10.0000	0.8549	0.0454
6	100.0000	0.8989	0.0342
7	100.0000	0.8396	0.0572
8	100.0000	0.8407	0.0530
9	100.0000	0.8560	0.0435
10	100.0000	0.8538	0.0505
11	100.0000	0.7967	0.0547
12	100.0000	0.8253	0.0394
13	100.0000	0.8253	0.0394
14	1.0000	0.8407	0.0697
15	100.0000	0.8549	0.0454
16	100.0000	0.8407	0.0278
17	100.0000	0.8253	0.0394
18	100.0000	0.8846	0.0337
19	1000.0000	0.8253	0.0394
20	10.0000	0.8407	0.0530

The search grids are finite: best means best among the stated candidates. Full CV results and final class coefficients are saved as CSVs. No CV candidate or final selected model emitted a convergence warning.

2(c)(i) Multiclass confusion matrix and ROC





Confusion-matrix rows are true activities, columns predicted activities. One-vs-rest ROC treats each activity as positive against all other activities. Every class has positive and negative test examples, so all seven curves are available. AUC measures ranking, not the accuracy of the multiclass argmax decision.

class	test_support	ovr_auc
bending1	2	1.0000
bending2	2	0.9706
cycling	3	1.0000
lying	3	1.0000
sitting	3	0.9167
standing	3	1.0000
walking	3	1.0000

$$AUC_{macro} = \frac{1}{7}\sum_{k=1}^{7} AUC_k, \qquad AUC_{weighted} = \sum_{k=1}^{7} \frac{n_k}{19} AUC_k$$

Macro AUC = 0.9839; weighted AUC = 0.9837. Macro gives each activity equal weight; weighted AUC uses test support. Errors: bending2 to sitting: 1; sitting to bending1: 1; sitting to standing: 1.

2(c)(ii) Gaussian and Multinomial Naive Bayes

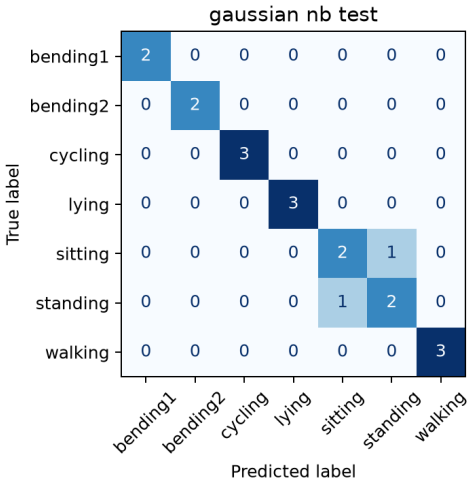
Both models search l = 1,...,20 with the same five multiclass folds. GaussianNB searches variance smoothing in {1e-9, 1e-6, 1e-3, 0.1, 1}; StandardScaler is fitted inside each fold. It models each feature as conditionally Gaussian and independent given activity.

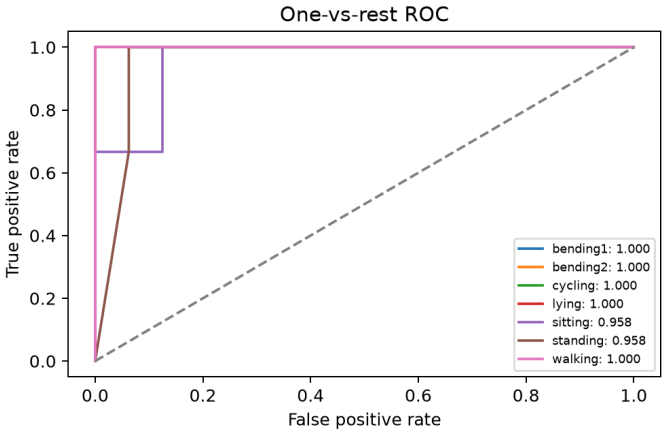
MultinomialNB searches alpha in {0.001, 0.01, 0.1, 1, 10}. All raw statistics are nonnegative. A within-fold MaxAbsScaler rescales columns without subtraction, so validation and test values remain nonnegative (they may exceed one). The features are continuous summaries, not counts, so the multinomial likelihood is only an approximation. Both models also simplify the strong dependence among summaries and adjacent segments.

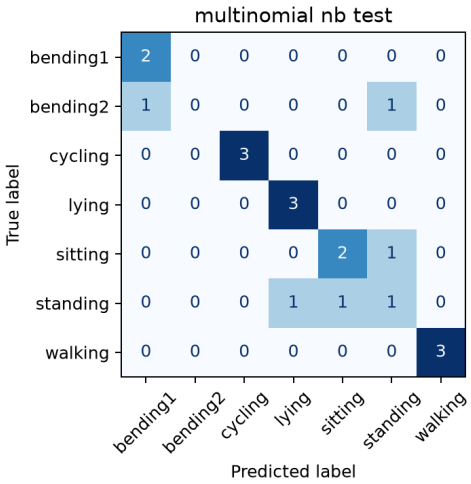
l	GNB smoothing	GNB CV acc.	MNB alpha	MNB CV acc.
1	0.1000	0.8967	0.0010	0.7385
2	0.0010	0.8846	0.0010	0.6659
3	0.1000	0.8681	1.0000	0.7088
4	0.1000	0.8407	0.0010	0.7231
5	0.1000	0.8835	0.1000	0.7824
6	0.1000	0.8681	1.0000	0.7527
7	0.1000	0.8692	0.1000	0.7516
8	0.1000	0.8978	0.1000	0.7802
9	0.1000	0.8407	0.1000	0.7670
10	0.1000	0.8549	0.1000	0.8110
11	0.1000	0.8549	0.1000	0.7802
12	0.1000	0.8396	0.1000	0.7813
13	0.1000	0.8264	0.1000	0.7813
14	0.1000	0.8253	0.0100	0.7956
15	0.1000	0.8264	0.0100	0.7670
16	0.1000	0.8407	0.0010	0.7802
17	0.1000	0.8396	0.0010	0.7967
18	0.1000	0.8121	0.0010	0.7824
19	0.1000	0.8407	0.0100	0.7824
20	0.1000	0.8253	0.1000	0.8110

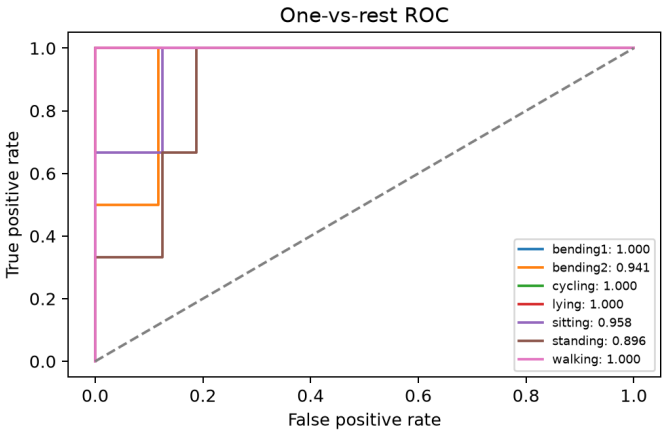
GaussianNB selects l = 8, smoothing = 0.1; CV accuracy 0.8978. MultinomialNB selects l = 10, alpha = 0.1; CV accuracy 0.8110. No test scores selected these settings.

2(c)(ii) Naive Bayes test results









activity	support	Gaussian OvR AUC	Multinomial OvR AUC
bending1	2	1.0000	1.0000
bending2	2	1.0000	0.9412
cycling	3	1.0000	1.0000
lying	3	1.0000	1.0000
sitting	3	0.9583	0.9583
standing	3	0.9583	0.8958
walking	3	1.0000	1.0000

Gaussian accuracy/error = 0.8947/0.1053; macro/weighted AUC = 0.9881/0.9868. Multinomial accuracy/error = 0.7368/0.2632; macro/weighted AUC = 0.9708/0.9708.

2(c)(iii) Compare the multiclass models

method	l	CV acc.	test acc.	test error	macro AUC
multinomial_l1	2	0.9429	0.8421	0.1579	0.9839
gaussian_nb	8	0.8978	0.8947	0.1053	0.9881
multinomial_nb	10	0.8110	0.7368	0.2632	0.9708

The CV winner(s): multinomial_l1. The test winner(s): gaussian_nb. With few recordings, model assumptions, variance smoothing, and the number of fitted parameters can strongly affect these comparisons. This does not prove that any distributional assumption is exactly true.

Gaussian errors: sitting to standing: 1; standing to sitting: 1. MultinomialNB errors: bending2 to bending1: 1; bending2 to standing: 1; sitting to standing: 1; standing to lying: 1; standing to sitting: 1. The multinomial count likelihood is a weaker match to continuous statistics. Logistic regression can model discriminative combinations, but many correlated segment features and few bending samples make estimation difficult.

All three methods use the same test recordings and multiclass folds. One test observation changes accuracy by 0.0526, so these results support a comparison on this split rather than a universal ranking. AUC can be high even when competing class probabilities lead to incorrect argmax labels.

3. ISLR 4.8.3: why the Bayes rule is quadratic

The exercise assumes one feature, K classes, and X given class k normally distributed with class-specific mean and variance. Let πk be the prior probability of class k. Bayes chooses the class with the largest posterior:

$$P(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_j \pi_j f_j(x)}$$

The denominator is common to all classes. Taking logs and dropping only the common -0.5 log(2π) gives:

$$\delta_k(x) = \log \pi_k - \log \sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2}$$

$$= -\frac{x^2}{2\sigma_k^2} + \frac{\mu_k}{\sigma_k^2}x - \frac{\mu_k^2}{2\sigma_k^2} + \log \pi_k - \log \sigma_k$$

For classes k and j, solve δk(x) - δj(x) = 0. Its quadratic coefficient is:

$$\frac{1}{2\sigma_j^2} - \frac{1}{2\sigma_k^2}$$

This coefficient generally does not cancel when variances differ, so the boundary comparison is quadratic. In one dimension it can give two crossing points. If the variances happen to be equal, the quadratic terms cancel and the comparison becomes linear. Thus QDA allows nonlinear boundaries, but special cases can be linear.

4. ISLR 4.8.7: probability of a dividend

The exercise gives mean profit 10 for dividend-paying companies and 0 for companies not paying dividends. Both groups have variance 36, so the standard deviation is 6. The priors are 0.8 and 0.2. For X = 4:

$$f_{Yes}(4) = \frac{1}{6\sqrt{2\pi}} \exp\left[-\frac{(4-10)^2}{72}\right]$$

$$f_{No}(4) = \frac{1}{6\sqrt{2\pi}} \exp\left[-\frac{(4-0)^2}{72}\right]$$

Bayes' rule uses both the density at the observed profit and the prior probability of each class:

$$P(Yes \mid X = 4) = \frac{0.8 f_{Yes}(4)}{0.8 f_{Yes}(4) + 0.2 f_{No}(4)}$$

$$= \frac{1}{1 + 0.25 e^{5/18}}$$

The common density constant cancels. The density ratio is fNo(4)/fYes(4) = exp((36 - 16)/72) = exp(5/18). The calculation below uses unrounded densities:

fYes(4)	fNo(4)	P(Yes | X=4)
0.0403	0.0532	0.7519

The estimated probability is 0.75185245, or about 75.19%. With a 0.5 threshold the prediction is Yes. Although profit 4 is closer to the non-dividend mean, the much larger dividend prior makes Yes more likely.

References

[1] Supplied Homework4.pdf, Questions 2-4. Supplied ISLP textbook: Exercise 4.8.3, printed p.193 (PDF p.202); Exercise 4.8.7, printed p.195 (PDF p.204).
[2] HW3 repository: github.com/jay-zhao1/DSCI552-HW3.
[3] Scikit-learn 1.8 documentation: logistic regression, cross-validation, Naive Bayes.
[4] Firth (1993), Bias reduction of maximum likelihood estimates, Biometrika 80(1), 27-38, doi:10.1093/biomet/80.1.27. logistf package documentation for Jeffreys-penalized logistic inference.

AI tools assisted with chapter concepts, unfamiliar code, and preparing code and explanations.